Web3User Permission Lab

Engineering Status & Development Roadmap

Status snapshot: 3 November 2026
Development branch: `feature/m13-d-consensus`
Development HEAD: `25bfe62fbb1542e2f8512f522f9c8e4892b4ad97`
Public release tag: `v0.1.0-rc.1` — historical release candidate, not the current development HEAD.

Executive status

Web3User has advanced beyond the M6-era public documentation. The repository history contains implementation milestones through M13-C.1 on `main`; the M13-D durable-witness / consensus foundation is committed on the development branch. The DR-2B Linux container verification harness is accepted, and Phase 1-A source protection has been committed and pushed.

The Cloudflare website is a static project-information preview. It is not a hosted production authorization backend, wallet, transaction broadcaster, or live consensus service.

Latest verification evidence

- **32 Linux-compatible test files passed.**
- **655 / 655 tests passed; zero failures.**
- 26 Windows real-etcd tests were excluded from the Linux container run and require separate native Windows validation.
- Basic authorization example, persistent replay demo, and npm package dry-run passed.
- PostgreSQL test service used ephemeral tmpfs storage and no published host port. The Docker bridge still permits outbound connectivity; it is not air-gapped.
- Phase 1-A commit: `25bfe62fbb1542e2f8512f522f9c8e4892b4ad97`; exactly five approved Docker / CI / documentation files were pushed.
- The public v0.1.0-rc.1 tag remains historical; no stable production release is claimed.

Development roadmap

Milestone	Status
M0–M8: evaluator, signing, persistence, budgets, and hardening	Implemented
M9–M10: adversarial and disaster-recovery hardening	Implemented in milestone history
M11: monotonic witness adapter and fail-closed recovery	Committed
M12 / M13-B: durable witness and bounded reconciliation pipeline	Committed
M13-C / C.1: durable witness lifecycle integration and startup synchronization remediation	Committed on main
M13-D: durable witness / consensus foundation	Committed on development branch
M13-D real-etcd integration harness	Pending report review and acceptance

DR-2B: non-root Docker + isolated Compose verification	Accepted; 655/655 Linux-compatible tests
Phase 1-A: source protection	Complete; commit pushed
Production readiness	Blocked by open engineering and operational gates
Stable public release	Not released

Production-readiness blockers

1. Revocation freshness: the 30-second fail-closed freshness gate still needs reliable background synchronization and integration validation.
2. Receipt canonicalization: gRPC and Engine currently use incompatible signing-preimage formats; unify the format and add verification tests.
3. Hardware-backed signing: AWS KMS Witness Signer adapter and algorithm / encoding contract tests remain outstanding.
4. Offline recovery: air-gapped backup-key ceremony and offline custody are not complete.
5. Cross-store recovery: PostgreSQL / etcd / SQLite backup and restore reconciliation is not proven end-to-end.
6. Authority and availability: PostgreSQL remains the live authority / single point of failure; etcd is not confirmed as the live authorization path.
7. RPO / RTO: no repeatable restore drill has demonstrated measured recovery objectives.
8. Independent security review: no independent commercial security audit has been completed.

Next sequence

1. Review and accept or reject the dedicated M13-D real-etcd harness report.
2. Run the 26 Windows real-etcd tests natively.
3. Fix revocation freshness and receipt-preimage compatibility.
4. Implement and contract-test KMS signing.
5. Complete offline key custody and cross-store restore exercises.
6. Measure RPO / RTO.
7. Obtain independent security review before production use.

Status discipline: Committed means present in Git history. Accepted means the stated validation evidence was reviewed. Production-ready requires all production blockers and operational gates to be closed.